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Robot HAT Hardware Components Documentation

The Robot HAT gives the Raspberry Pi more control over the hardware used in robotics projects. It connects motors, servos, sensors, audio, communication devices, and battery power. These features help the Raspberry Pi interact with the physical parts that let a robot sense its surroundings and move. In a compatible setup such as PiDog V2, the Robot HAT connects software processing with robotic actions.

For motor and servo control, the Robot HAT provides a built-in motor driver with two DC motor channels that use direction and PWM signals. PiDog does not use DC motors for walking because its movement comes from servos. The board provides 12 PWM channels for servo control, which handle PiDog's leg, head, and tail movement (SunFounder, "Robot HAT"; SunFounder, "PiDog"). It also includes four 12-bit ADC channels for analog sensor values and digital connections that can support the ultrasonic sensor. The documentation does not list a general-purpose DAC output, but the Robot HAT provides I2S digital audio output and a mono speaker for sound support (SunFounder, "Robot HAT").

For communication with sensors and other modules, the Robot HAT provides I²C, UART, and SPI expansion ports. Its 6.0–8.4 V power input can power both the Raspberry Pi and Robot HAT, while the USB-C port allows the battery to charge (SunFounder, "Robot HAT"). An ultrasonic sensor can connect through the Robot HAT's digital pins to measure distance and help a robot detect nearby obstacles (SunFounder, "Robot HAT"). Two power indicators show the board's power status, while the board also includes a user LED that provides extra status feedback. For computer vision, PiDog's camera module connects directly to the Raspberry Pi through its CSI, or Camera Serial Interface, port rather than through the Robot HAT (SunFounder, "PiDog").

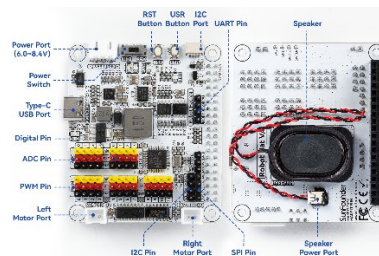


Figure 1 Robot HAT Board with Main Ports and Connectors (SunFounder)

PiDog V2 shows how these components work together in a real robotics application (Figure 2). Its ultrasonic module detects obstacles ahead, while the camera supports color recognition and face detection. PiDog also includes touch sensors, a sound-direction sensor, and

a 6-DOF IMU, while its 12 servos create movement (SunFounder, “PiDog”). For example, the ultrasonic sensor can detect a nearby object, the Raspberry Pi can process the distance, and the servos can adjust PiDog’s movement in response. The V2 version also updates the Robot HAT and servo configuration to support Raspberry Pi 5.

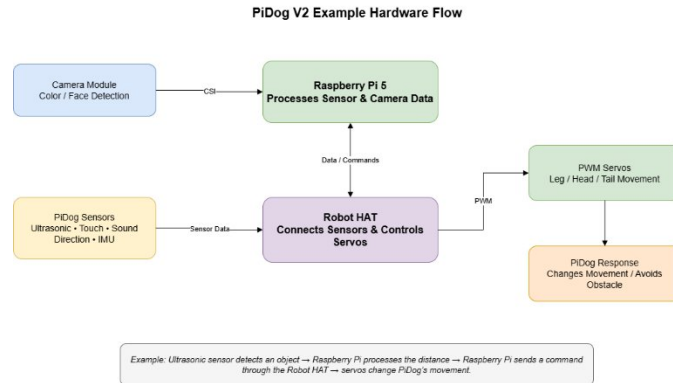


Figure 2 Simplified Robot HAT Hardware Flow (draw.io)

Power, wiring, and compatibility require attention when using the Robot HAT. Motors and several servos can draw a large amount of current, so the robot needs a stable battery and secure power connections. Users must also connect sensors, motors, and communication devices to the correct ports because incorrect wiring can prevent the system from working properly. Compatibility matters because the standard PiDog version does not support Raspberry Pi 5, while PiDog V2 includes updated hardware that does (SunFounder, “PiDog”). Because of this, checking the wiring and hardware before testing can help prevent setup issues.

In conclusion, the Robot HAT expands the Raspberry Pi by adding the connections and control features needed for robotics. Its motor driver, PWM servo channels, ultrasonic support, ADC inputs, audio output, communication interfaces, battery system, and LED indicators connect computing with physical hardware. PiDog V2 shows how these features can support sensing, movement, computer vision, sound, and interaction in one system. With correct wiring, stable power, and compatible hardware, the Robot HAT can provide a practical platform for robotics and AI projects.

References

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